



Institute of
Information Technology

Intrusion Detection System and Intrusion Prevention System

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Laboratory



Institute of
Information Technology

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► Course:

- **Practice:** IDS/IPS
- **Level:** Bachelor
- **Major:** Cyber Security



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LAB WORK 01

Create and Set Up a Virtual Lab Environment

Virtualization, network interfaces, and isolated
lab networks

All Slide Shows: <https://cygnussec.com/project/ids-ips>



Learning Foundations

SOFT SKILLS

- Use hypervisor tools such as VirtualBox, VMware Workstation, ESXi, and Hyper-V.
- Apply basic networking knowledge to an isolated virtual lab.
- Operate Linux systems confidently from the command line.



Prepare the Lab Components

- Download and install a virtual machine tool.
- Obtain the operating-system images or virtual appliances:
 - Kali Linux or another attack machine
 - Ubuntu Server as the router
 - Metasploitable2 as the victim machine



Choose the Virtualization Stack

VIRTUAL MACHINE TOOLS

- VirtualBox
- VMware Workstation



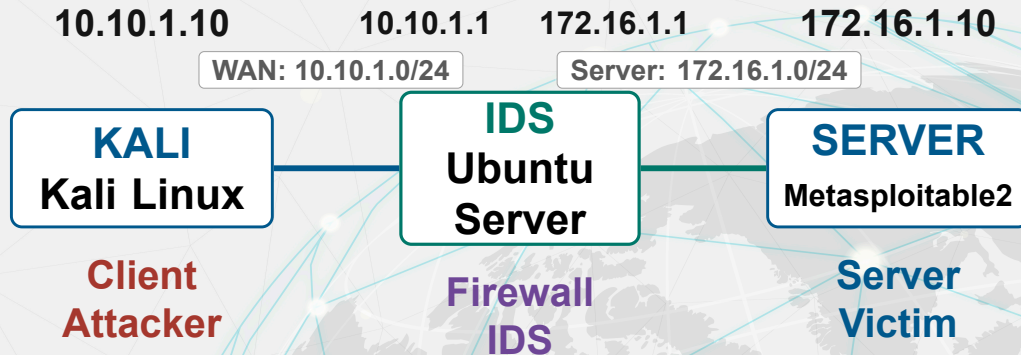
Choose the Guest Systems

OPERATING SYSTEMS

- Kali Linux: attacker
- Ubuntu Server: router
- Metasploitable2: victim



Lab 1 Topology





Build the Environment in Five Steps

1. Install the virtual machine tool.
2. Install or import each operating system.
3. Create and attach the virtual networks.
4. Configure networking inside each guest OS.
5. Test connectivity and validate the topology.



Allocate Resources to Each Machine

- Install the hypervisor and create the three virtual machines.
- Suggested resource allocation:
 - Attacker: 3 vCPU, 4 GB RAM, 80 GB storage
 - Router: 2 vCPU, 3 GB RAM, 50 GB storage
 - Victim: 1 vCPU, 512 MB RAM, 8 GB storage



Create Two Isolated Virtual Networks

SERVER NETWORK

172.16.1.0/24

Victim-facing segment.

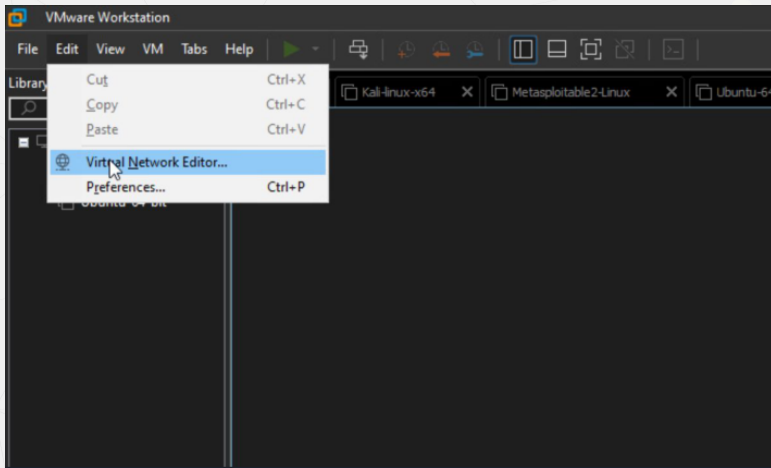
WAN NETWORK

10.10.1.0/24

Attacker-facing segment.

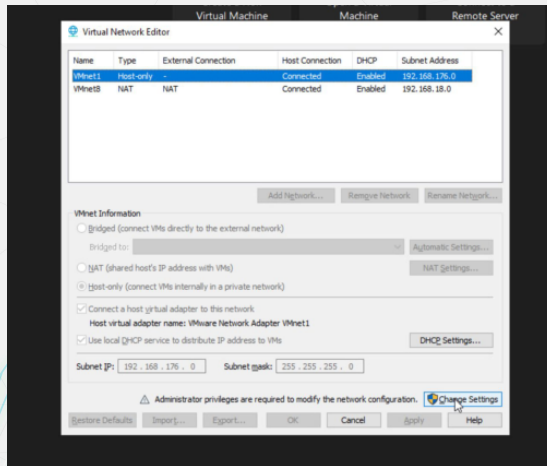


Open VMware Virtual Network Editor





Review the Existing VMware Networks





Create the Server Network

Virtual Network Editor

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	Bridged	Auto-bridging	-	-	-
VMnet1	Host-only	-	Connected	Enabled	192.168.176.0
VMnet8	NAT	NAT	Connected	Enabled	192.168.18.0
Server4...	Host-only	-	Connected	Enabled	172.16.1.0

Add Network... Remove Network Rename Network...

VMnet Information

☐ Bridged (connect VMs directly to the external network)
Bridged to: Automatic Automatic Settings...

☐ NAT (shared host's IP address with VMs) NAT Settings...

☒ Host-only (connect VMs internally in a private network)

☒ Connect a host virtual adapter to this network
Host virtual adapter name: VMware Network Adapter VMnet2

☒ Use local DHCP service to distribute IP address to VMs DHCP Settings...

Subnet IP: 172.16.1.0 Subnet mask: 255.255.255.0

Restore Defaults Import... Export... OK Cancel Apply Help



Verify Both VMware Networks

Virtual Network Editor

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	Bridged	Auto-bridging	-	-	-
VMnet1	Host-only	-	Connected	Enabled	192.168.176.0
ServerNetwork	Host-only	-	Connected	Enabled	172.16.1.0
WANNetwork	Host-only	-	Connected	Enabled	10.10.1.0
VMnet8	NAT	NAT	Connected	Enabled	192.168.18.0

VMnet Information

☐ Bridged (connect VMs directly to the external network)
Bridged to: Automatic Automatic Settings...

☐ NAT (shared host's IP address with VMs)
NAT Settings...

☒ Host-only (connect VMs internally in a private network)

☒ Connect a host virtual adapter to this network
Host virtual adapter name: VMware Network Adapter VMnet2

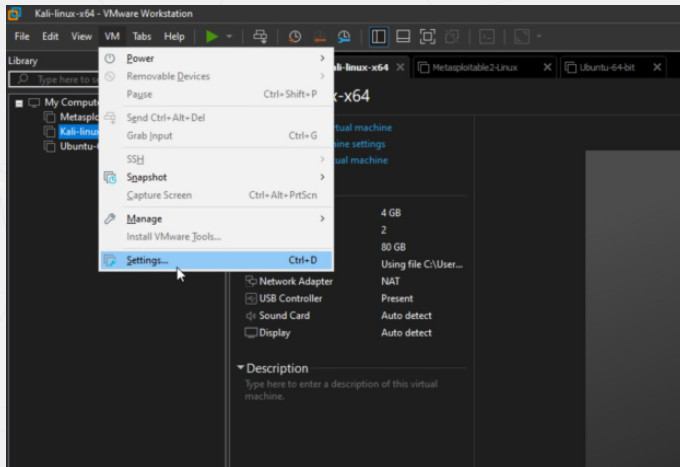
☒ Use local DHCP service to distribute IP address to VMs
DHCP Settings...

Subnet IP: 172.16.1.0 Subnet mask: 255.255.255.0

Restore Defaults Import... Export... OK Cancel Apply Help

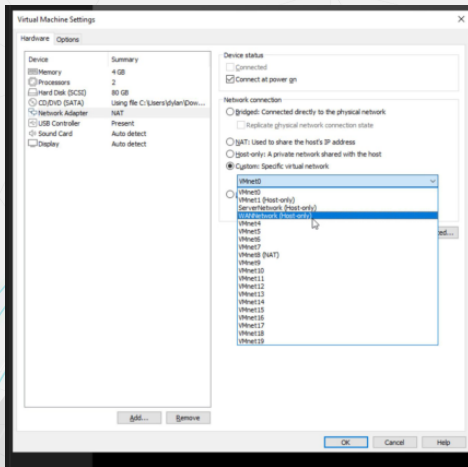


Open VMware Network Settings



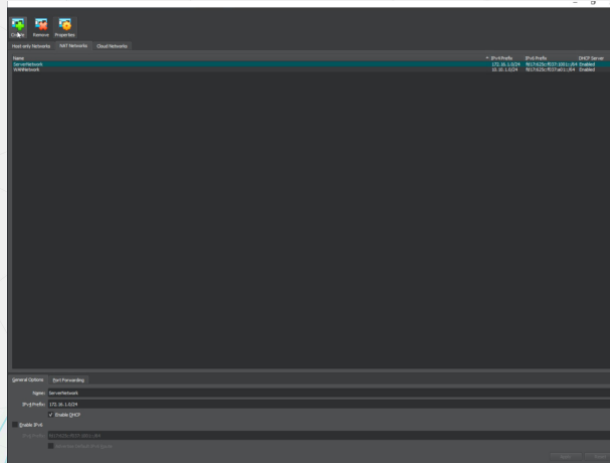


Attach Each VMware Adapter to the Correct Network



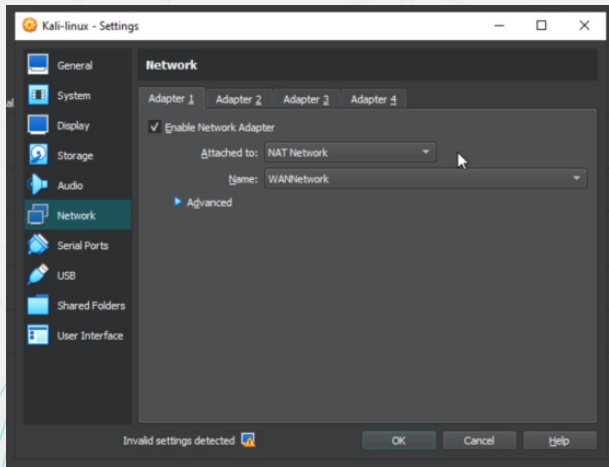


Create the Networks in VirtualBox





Attach a VirtualBox Adapter to the WAN Network





Configure Ubuntu's WAN Interface

WAN: ens160

```
ip link set dev ens160 down
```

```
ip addr add 10.10.1.1/24 dev ens160
```

```
ip link set dev ens160 up
```

```
ip route
```



Configure Ubuntu's Server Interface

SERVER: ens192

```
ip link set dev ens192 down
```

```
ip addr add 172.16.1.1/24 dev ens192
```

```
ip link set dev ens192 up
```

Verify both connected routes and their source addresses.



Make the Ubuntu Network Configuration Persistent

Edit `/etc/netplan/00-installer-config.yaml`:

```
network:
  ethernets:
    ens160:
      addresses: [10.10.1.1/24]
      dhcp4: false
```



Add the Server-Facing Interface to Netplan

Continue the **ethernets** configuration:

```
ens192:  
  addresses: [172.16.1.1/24]  
  dhcp4: false  
version: 2
```

Reference: [Netplan configuration examples](#)



Enable IPv4 Forwarding Persistently

Ensure `net.ipv4.ip_forward` is enabled in `/etc/sysctl.conf`:

```
net.ipv4.ip_forward=1  
sudo sysctl -p /etc/sysctl.conf
```

The Ubuntu router must forward packets between the WAN and server networks.



Temporary IPv4 Forwarding

Use either command-line method for a temporary setting:

```
sudo sysctl -w net.ipv4.ip_forward=1
```

or

```
P=/proc/sys/net/ipv4/ip_forward  
sudo echo 1 > $P
```



Configure Kali and Metasploitable2 Temporarily

Replace the placeholders with the host's assigned address and router gateway.

```
ip link set dev eth0 down
```

```
ip addr add x.x.x.x/24 dev eth0
```

```
ip link set dev eth0 up
```

```
ip route add default via x.x.x.1
```

ip route Confirm the address, connected route, and default route.



Persist the Kali and Metasploitable2 Configuration

Edit a file under `/etc/network/interfaces.d/*`:

```
allow-hotplug eth0  
iface eth0 static  
    address 10.10.1.x/24  
    gateway 10.10.1.1
```



Apply the Kali and Metasploitable2 Configuration

Apply the configuration: `sudo systemctl restart networking`

Reference: [Debian network configuration](#)



Validate Connectivity with Ping

`ping 10.10.1.1` Client-side gateway

`ping 172.16.1.1` Server-side gateway

`ping 10.10.1.x` Attacker host

`ping 172.16.1.x` Victim host



Pull the Lab Images from the Registry

```
REG=registry.cygnussec.tech/labs  
docker login registry.dylantran.tech  
docker pull $REG/kali-linux:latest  
docker pull $REG/ubuntu:22.04  
docker pull $REG/metasploitable2:latest
```

Use the course-provided registry credentials; do not store passwords in the slide source.



Create the WAN Docker Network

```
D="docker network create"
```

```
N=10.10.1.0/24
```

```
R="--ip-range=$N"
```

```
S="--subnet=$N"
```

```
G="--gateway=10.10.1.254"
```

```
$D $R $S $G WANNetwork
```



Create the Server Docker Network

```
D="docker network create"  
N=172.16.1.0/24  
R="--ip-range=$N"  
S="--subnet=$N"  
G="--gateway=172.16.1.254"  
$D $R $S $G ServerNetwork
```



Start the Attacker Container

```
RUN="docker run -dit --privileged"  
NET="--network WANNetwork"  
HOST="--hostname kali -p 9392:9392"  
IMAGE=$REG/kali-linux:latest  
$RUN $NET $HOST --name attacker $IMAGE
```



Start the Router Container

```
RUN="docker run -dit --privileged"  
NET="--network WANNetwork"  
HOST="--hostname ubuntu --name router"  
IMAGE=$REG/ubuntu:22.04  
$RUN $NET $HOST $IMAGE  
SN=ServerNetwork  
docker network connect $SN router
```



Start the Victim Container

```
RUN="docker run -dit --privileged"  
NET="--network ServerNetwork"  
HOST="--hostname msfadmin --name victim"  
IMAGE=$REG/metasploitable2:latest  
$RUN $NET $HOST $IMAGE  
docker attach <container_name>
```



Download the Required Software

- VirtualBox: <https://www.virtualbox.org/>
- VMware: <https://www.vmware.com/>
- Kali Linux: <https://www.kali.org/>
- Ubuntu Server:
<https://ubuntu.com/download/server>
- Metasploitable2: <https://sourceforge.net/projects/metasploitable/>



Complete the Lab by Validating Every Path

1. Confirm that every machine is attached to the intended virtual network.
2. Verify IP addresses, routes, and IPv4 forwarding.
3. Ping both router interfaces and the remote host.
4. Record the final topology and validation results.